

E-commerce Data Analysis

Query 1: List all unique cities where customers are located:

SQLQuery2.sql - LA...R1AT2P\tapas (51))

SQLQuery1.sql - LA...R1AT2P\tapas (70))*

```
select distinct customer_city
from customers
```

200 %

Results Messages

	customer_city
1	claudia
2	santa salete
3	sao jose dos cordeiros
4	santa izabel do oeste
5	andradina
6	mostardas
7	buritis
8	cacule
9	santa quiteria
10	novo jardim
11	sud mennucci
12	deodapolis
13	lajeado
14	santo antonio de posse
15	planalto
16	quinta do sol
17	jardim do serido
18	santanesia
19	oliveira dos brejinhos
20	sao goncalo do abaete
21	goianinha
22	aluminio
23	itauna
24	santa vitoria do palmar
25	santo antonio do rio a...
26	altos
27	erico cardoso
28	jaguarembé
29	rio espera
30	agrolandia
31	uirana
32	cerrito
33	laranjeiras do sul
34	altaneira
35	itapui
36	rio casca
37	belo vale

Query 2: Count the number of orders placed in 2017

```
select count(order_id) as orders_in_2017
from orders
where year(order_purchase_timestamp) = 2017
```

150 %

Results Messages

	orders_in_2017
1	45101

Query 3: Find the total sales per category

```
select products.product_category,
round(sum(payments.payment_value),2) sales
from products join order_items
on products.product_id = order_items.product_id
join payments
on payments.order_id = order_items.order_id
group by products.product_category
order by sales desc
```

150 %

Results Messages

	product_category	sales
1	bed table bath	1712553.67
2	HEALTH BEAUTY	1657373.12
3	computer accessories	1585330.45
4	Furniture Decoration	1430176.39
5	Watches present	1429216.68
6	sport leisure	1392127.56
7	housewares	1094758.13
8	automotive	852294.33
9	Garden tools	838280.75
10	Cool Stuff	779698
11	Furniture office	646826.49
12	toys	619037.69
13	babies	539845.66
14	perfumery	506738.66
15	telephony	486882.05
16	stationary store	317440.07
17	pet Shop	311268.97
18	PCs	279121.55
19	electronics	259857.1
20	NULL	252801.71
21	Construction Tools Construction	241475.63
22	musical instruments	233074.12
23	electrostile	225584.38
24	Fashion Bags and Accessories	218158.28
25	fixed telephony	207010.26
26	Games consoles	195480.38
27	Bags Accessories	187151.29
28	Casa Construc�o	136645.29
29	Room Furniture	136138.77

Query 4: Count the number of customers from each state

```
select customer_state , count(customer_id) as no_of_customers
from customers
group by customer_state
order by no_of_customers desc
```

150 %

Results Messages

	customer_state	no_of_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020
11	PE	1652
12	CE	1336
13	PA	975
14	MT	907
15	MA	747
16	MS	715
17	PB	536
18	PI	495
19	RN	485
20	AL	413
21	SE	350
22	TO	280
23	RO	253
24	AM	148
25	AC	81
26	AP	68
27	RR	46

Query 5: Calculate the number of orders per month in 2018

```
select month(order_purchase_timestamp) months, count(order_id) order_count
from orders where year(order_purchase_timestamp) = 2018
group by month(order_purchase_timestamp)
order by month(order_purchase_timestamp)
```

150 %

Results Messages

	months	order_count
1	1	7269
2	2	6728
3	3	7211
4	4	6939
5	5	6873
6	6	6167
7	7	6292
8	8	6512
9	9	16
10	10	4

Query 6: Find the average number of orders per product, grouped by customer city

```
with count_per_order as
(
select orders.order_id, orders.customer_id, count(order_items.order_id) as oc
from orders join order_items
on orders.order_id = order_items.order_id
group by orders.order_id, orders.customer_id)

select customers.customer_city, round(avg(count_per_order.oc),2) average_orders
from customers join count_per_order
on customers.customer_id = count_per_order.customer_id
group by customers.customer_city order by average_orders desc
```

150 %

Results Messages

	customer_city	average_orders
1	padre carvalho	7
2	datas	6
3	candido godoi	6
4	celso ramos	6
5	matias olimpio	5
6	cidelandia	4
7	morro de sao paulo	4
8	picarra	4
9	teixeira soares	4
10	curralinho	4
11	pacuja	3
12	capela	3
13	alto paraíso de goias	3
14	inconfidentes	3
15	pedregulho	3
16	nova esperanca do sul	3
17	brasileia	3
18	ipua	3
19	ubata	3
20	jose boiteux	3
21	buniti	3
22	ouvidor	3
23	chapadao do lageado	3
24	pau brasil	2

Query 7: Calculate the percentage of total revenue contributed by each product category

```
select products.product_category,  
round((sum(payments.payment_value)/(select sum(payment_value) from payments))*100,2) sales_percentage  
from products join order_items  
on products.product_id = order_items.product_id  
join payments  
on payments.order_id = order_items.order_id  
group by products.product_category  
order by sales_percentage desc
```

150 %

Results Messages

	product_category	sales_percentage
1	bed table bath	10.7
2	HEALTH BEAUTY	10.35
3	computer accessories	9.9
4	Watches present	8.93
5	Furniture Decoration	8.93
6	sport leisure	8.7
7	housewares	6.84
8	automotive	5.32
9	Garden tools	5.24
10	Cool Stuff	4.87
11	Furniture office	4.04
12	toys	3.87
13	babies	3.37
14	perfumery	3.17
15	telephony	3.04
16	stationary store	1.98
17	pet Shop	1.94
18	PCs	1.74
19	electronics	1.62
20	NULL	1.58
21	Construction Tools Construction	1.51
22	musical instruments	1.46
23	electrostyle	1.41
24	Fashion Bags and Accessories	1.36
25	fixed telephony	1.29
26	Games consoles	1.22
27	Bags Accessories	1.17
28	Room Furniture	0.85
29	Casa Construção	0.85

Query 8: Calculate the total revenue generated by each seller, and rank them by revenue

```
select *, dense_rank() over(order by revenue desc) as rn from
(select order_items.seller_id, sum(payments.payment_value) revenue
from order_items
join payments on order_items.order_id = payments.order_id
group by order_items.seller_id) as a
```

150 %

Results Messages

	product_category	sales_percentage
1	bed table bath	10.7
2	HEALTH BEAUTY	10.35
3	computer accessories	9.9
4	Watches present	8.93
5	Furniture Decoration	8.93
6	sport leisure	8.7
7	housewares	6.84
8	automotive	5.32
9	Garden tools	5.24
10	Cool Stuff	4.87
11	Furniture office	4.04
12	toys	3.87
13	babies	3.37
14	perfumery	3.17
15	telephony	3.04
16	stationary store	1.98
17	pet Shop	1.94
18	PCs	1.74
19	electronics	1.62
20	NULL	1.58
21	Construction Tools Construction	1.51
22	musical instruments	1.46
23	electrostile	1.41
24	Fashion Bags and Accessories	1.36
25	fixed telephony	1.29
26	Games consoles	1.22
27	Bags Accessories	1.17
28	Room Furniture	0.85
29	Casa Construc�o	0.85
30	ELECTRICES 2	0.78
31	Agro Industria e Comercio	0.74
32	home appliances	0.6
33	climatization	0.57
34	House comfort	0.53
35	Furniture Kitchen Service Are...	0.47

Query 9: Calculate the moving average of order values for each customer over their order history

```
select customer_id, order_purchase_timestamp, payment,
avg(payment) over(partition by customer_id order by order_purchase_timestamp
rows between 2 preceding and current row) as mov_avg
from
(select orders.customer_id, orders.order_purchase_timestamp,
payments.payment_value as payment
from payments
join orders on payments.order_id = orders.order_id) as a
```

150 %

Results Messages

	customer_id	order_purchase_timestamp	payment	mov_avg
1	00012a2ce6f8dcca20d059ce98491703	2017-11-14 16:08:26.0000000	114.74	114.74
2	000161a058600d5901f007fab4c27140	2017-07-16 09:40:32.0000000	67.41	67.41
3	0001fd6190edaaf84bca3d49edf079	2017-02-28 11:06:43.0000000	195.42	195.42
4	0002414f95344307404f0ace7a26f1d5	2017-08-16 13:09:20.0000000	179.35	179.35
5	000379cdec625522490c315e70c7a9fb	2018-04-02 13:42:17.0000000	107.01	107.01
6	0004164d20a9e969af783496f3408652	2017-04-12 08:35:12.0000000	71.8	71.8
7	000419c5494106c306a97b5635748086	2018-03-02 17:47:40.0000000	49.4	49.4
8	00046a560d407e99b969756e0b10f282	2017-12-18 11:08:30.0000000	166.59	166.59
9	00050bf6e01e69d5c0fd612f1bcb69c	2017-09-17 16:04:44.0000000	85.23	85.23
10	000598caf2ef4117407665ac33275130	2018-08-11 12:14:35.0000000	1255.71	1255.71
11	0005aefbb696d34b3424dcd0a0e9fd0	2018-06-20 09:46:53.0000000	147.33	147.33
12	00062b33cb9f6fe976afdcf967ea74d	2017-03-15 23:44:09.0000000	58.95	58.95
13	00066ccbe787a588c52bd5f404590e3	2018-02-06 16:10:09.0000000	270	270
14	00072d033fe2e59061ae5c3aff1a2be5	2017-09-01 09:24:39.0000000	106.97	106.97
15	0009a69b72033b2d0ec8c69fc70ef768	2017-04-28 13:36:30.0000000	173.6	173.6
16	000bf8121c3412d3057d32371c5d3395	2017-10-11 07:44:31.0000000	45.56	45.56
17	000e943451fc2788ca6ac98a682f2f49	2017-04-20 19:37:14.0000000	26.8	26.8
18	000e943451fc2788ca6ac98a682f2f49	2017-04-20 19:37:14.0000000	26.8	26.8
19	000e943451fc2788ca6ac98a682f2f49	2017-04-20 19:37:14.0000000	26.8	26.8
20	000e943451fc2788ca6ac98a682f2f49	2017-04-20 19:37:14.0000000	25.83	26.4766666666667
21	000f17e290c26b28549908a04cfe36c1	2017-11-10 16:37:05.0000000	139.52	139.52
22	000fd45dfedae68fc6676036610f879	2018-04-15 15:55:01.0000000	66.81	66.81
23	0010068a73b7c56da5758c39e5c7375	2017-04-11 18:19:55.0000000	54.83	54.83
24	001028b78fd413e19704b3867c369d3a	2018-04-12 12:51:28.0000000	37.61	37.61
25	00104a47c29da701ce41ee52077587d9	2018-01-14 20:28:56.0000000	100.34	100.34
26	001051abfcfdbed9f87b4266213a5df1	2018-05-30 09:19:31.0000000	13.35	13.35
27	001051abfcfdbed9f87b4266213a5df1	2018-05-30 09:19:31.0000000	32.03	22.69
28	001051abfcfdbed9f87b4266213a5df1	2018-05-30 09:19:31.0000000	19.82	21.7333333333333

Query executed successfully.

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Query 10: Calculate the cumulative sales per month for each year

```

select years, months , payment, sum(payment)
over(order by years, months) cumulative_sales from
(select year(orders.order_purchase_timestamp) as years,
month(orders.order_purchase_timestamp) as months,
round(sum(payments.payment_value),2) as payment from orders join payments
on orders.order_id = payments.order_id
group by year(orders.order_purchase_timestamp), month(orders.order_purchase_timestamp)) as a

```

	years	months	payment	cumulative_sales
1	2016	9	252.24	252.24
2	2016	10	59090.48	59342.72
3	2016	12	19.62	59362.34
4	2017	1	138488.04	197850.38
5	2017	2	291908.01	489758.39
6	2017	3	449863.6	939621.99
7	2017	4	417788.03	1357410.02
8	2017	5	592918.82	1950328.84
9	2017	6	511276.38	2461605.22
10	2017	7	592382.92	3053988.14
11	2017	8	674396.32	3728384.46
12	2017	9	727762.45	4456146.91
13	2017	10	779677.88	5235824.79
14	2017	11	1194882.8	6430707.59
15	2017	12	878401.48	7309109.07
16	2018	1	1115004.18	8424113.25
17	2018	2	992463.34	9416576.59
18	2018	3	1159652.12	10576228.71
19	2018	4	1160785.48	11737014.19
20	2018	5	1153982.15	12890996.34
21	2018	6	1023880.5	13914876.84
22	2018	7	1066540.75	14981417.59
23	2018	8	1022425.32	16003842.91
24	2018	9	4439.54	16008282.45
25	2018	10	589.67	16008872.12

Query 11: Calculate the year-over-year growth rate of total sales

```

with cte as
(
select year(orders.order_purchase_timestamp) as years,
round(sum(payments.payment_value),2) as payment from orders join payments
on orders.order_id = payments.order_id
group by year(orders.order_purchase_timestamp))

select years, ((payment-lag(payment,1) over(order by years))
/lag(payment,1) over(order by years))*100 YOY_growth
from cte

```

	years	YOY_growth
1	2016	NULL
2	2017	12112.7037613409
3	2018	20.0009238115826

Query 12: Identify the top 3 customers who spent the most money in each year

```
select years, customer_id, payment, d_rank from
(select YEAR(orders.order_purchase_timestamp) years, orders.customer_id, sum(payments.payment_value) payment,
DENSE_RANK() over(partition by YEAR(orders.order_purchase_timestamp) order by sum(payments.payment_value) desc) d_rank
from orders
join payments on payments.order_id = orders.order_id
group by YEAR(orders.order_purchase_timestamp), orders.customer_id) as a
where d_rank <= 3
```

130 %

Results Messages

	years	customer_id	payment	d_rank
1	2016	a9dc96b027d1252b0ac0a9b72d837c6	1423.55	1
2	2016	1d34ed25963d5aae4c3d73a4cda173	1400.74	2
3	2016	4a06381959b6670756de02e07b83815f	1227.78	3
4	2017	1617b1357756262b6a56ab541c47bc16	13664.08	1
5	2017	c6a2731c5b391845f6800c97401a43a9	6929.31	2
6	2017	3fd6777bbce08a352b0d04e4a7cc8f6	6726.66	3
7	2018	ec5b2ba62e574342386871631fad3fc	7274.88	1
8	2018	f48d464a0baaea338cb258f16991ab1f	6922.21	2
9	2018	e0a2412720e9ea428c1ac9896a7358	4809.44	3